Christian Ihejirika

Graduate Mechanical Engineer · Mini Design Portfolio

BEng Aerospace Engineering, University of Nottingham (2026) · cihejirika75@gmail.com · github.com/ChristianCI05

I work at the mechanical–electrical boundary and take hardware from concept through systems engineering into quantified, first-order analysis. Two projects follow. The first is a ground-up mechanical design study — a compact football launcher — carried from stakeholder needs to a 20,000-run feasibility model that is now steering the CAD. The second is a control and power-electronics simulation study from my final-year dissertation. Every figure and number here comes from my own models, and I'd be happy to walk through the assumptions behind them.

PROJECT 01 / MECHANICAL DESIGN & SYSTEMS ENGINEERING

Compact American-Football Launcher

Concept · full systems-engineering package · 20,000-run feasibility study complete · CAD & prototype next

A compact, single-operator launcher for players who want to train alone — architected as an automatable platform for programmable plays and performance assessment. It is designed to decouple ball speed from spin and to be more portable and lower-cost than bulky institutional (JUGS-style) machines, which assume a facility that already has people to throw.

Live repository (updated as the project progresses): github.com/ChristianCI05/Football-launcher-design-portfolio

Goal. Prove — before committing to CAD or hardware — that a single-operator, speed/spin-decoupled launcher is physically feasible, and find where it is weakest.

Approach. Systems engineering first: stakeholders → concept of operations → a 25-requirement product design specification with explicit pass/fail verification rules → an RFLP (requirements–functional–logical–physical) architecture. That framework then drives a first-order Python feasibility study; CAD and manufacture follow.

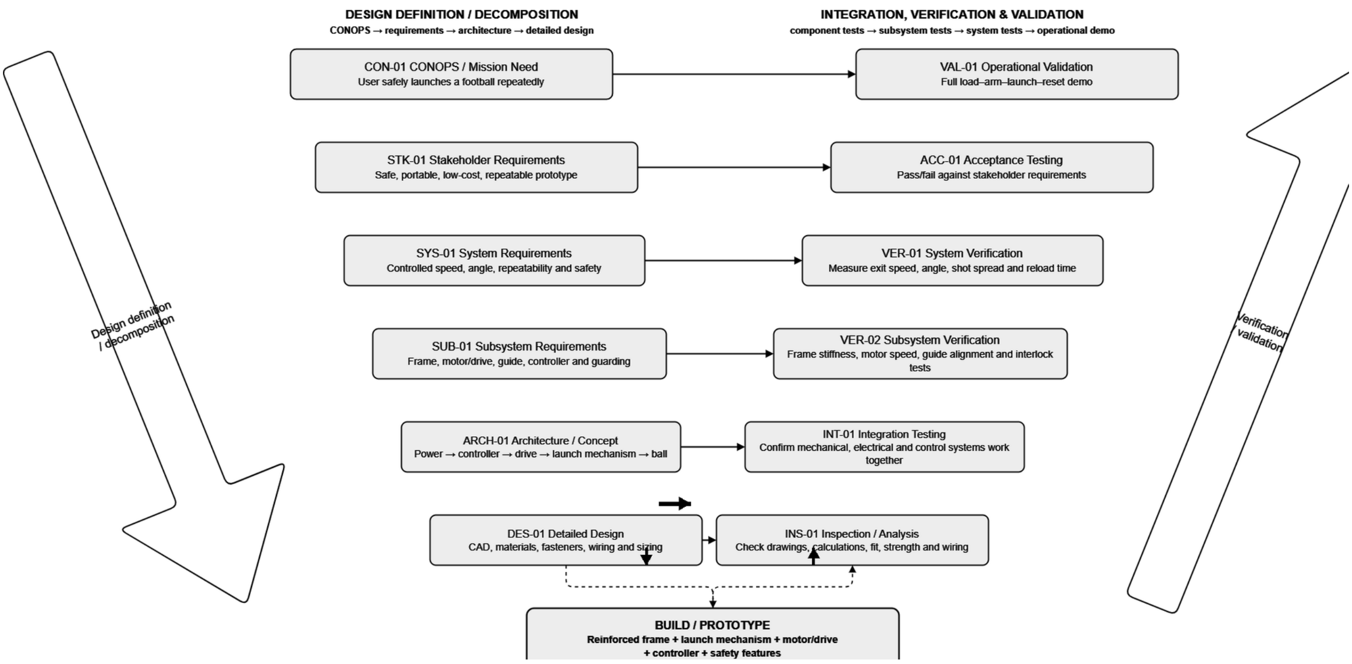

Fig. 1 — Systems-engineering V-model. Left leg decomposes mission need (CON-01) through stakeholder, system and subsystem requirements to detailed design; the right leg maps each level to its verification and validation activity. This traceability is the backbone the feasibility study and CAD hang off.

OPERATING CONCEPT

How the machine is meant to be used — and made safe

The concept of operations was written before any sizing: it defines what must be true for the machine to be usable by one person and safe around a rotating, energy-storing mechanism. Three ideas drive the sequence — a **guard interlock** (the spin motor cannot arm unless the loading guard is closed, and any out-of-range state drops to a safe-inhibit or fault branch); **speed/spin decoupling** (a pre-spun contactor set imparts spin while an elastic carriage sets exit velocity, so the two are tuned independently); and a **repeatable reset** that returns the system to a known state to keep shot-to-shot variation low.

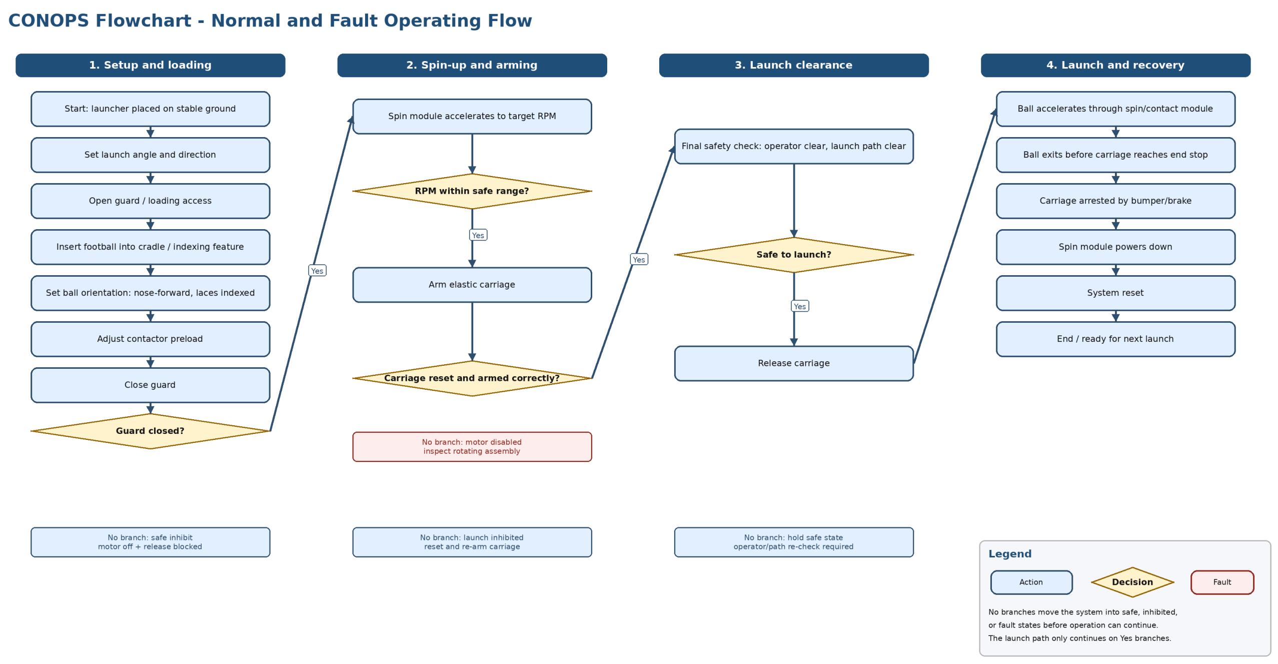

__Fig. 2 — CONOPS operating flow (normal and fault paths).__ The launch path only advances on the 'Yes' branch of each safety gate — guard closed, RPM in range, carriage armed, path clear. Every gate has a defined safe-inhibit or fault branch (see legend), so the machine fails to a safe state rather than firing in an unknown one. The equivalent state diagram is in the repository.

FEASIBILITY METHOD

With the operating concept fixed, a first-order physics model of the launch, spin and stability subsystems was wrapped in a **20,000-run Monte-Carlo** over **18 parameters** (masses, geometry, efficiency, friction, launch height). Each run is scored against the PDS pass/fail rules, and one-at-a-time tornado sensitivities rank the drivers. These are deliberately **first-order screening estimates** — they size the problem and locate the weaknesses that matter, not a validated performance prediction.

FEASIBILITY RESULTS

The launch subsystem is comfortably feasible

Across 20,000 sampled configurations the launch and spin subsystems sit in a healthy range. Ball kinetic energy centres on **~47 J** (19–90 J), from a stored input of **~95 J** median, at a peak launch force of **~276 N** nominal (P95 ≈ 650 N). The spin friction margin passes in **100%** of runs and required contactor speed (~2000 rpm nominal) passes in **99%**. With a **200 mm arrest stroke** the carriage-arrest force is fully mitigated (**100% pass**). Frame tip-over and sliding remain the only governing constraints.

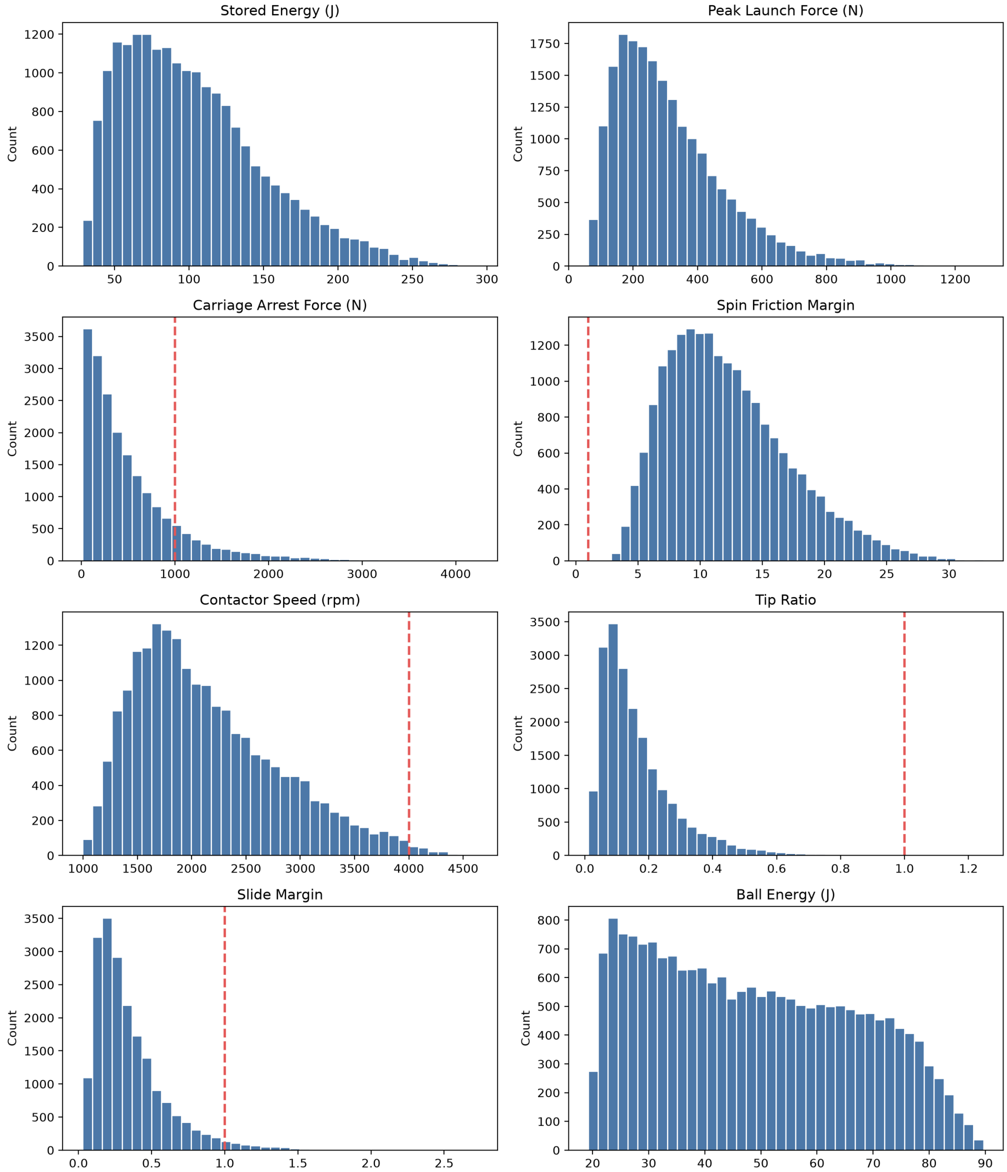

Fig. 3 — Monte-Carlo output distributions (20,000 runs). *Red dashed lines are PDS limits. Stored energy, launch force, spin margin and contactor speed all fall in workable bands, and the carriage-arrest screening is resolved by a 200 mm arrest stroke; tip-ratio and slide-margin (bottom, with limits far to the right) are the sole remaining constraints —* *see the next page.*

Verification check	Pass	Caution	Fail	Verdict
Spin friction margin	100%	0%	0%	**Robust**
Contactor speed (rpm)	99%	1%	0%	**Robust**
Carriage arrest force	100%	0%	0%	**Mitigated by 200 mm stroke**
Frame tip-over (tip ratio)	0%	0%	100%	**Governing constraint**
Frame sliding (slide margin)	0.3%	2.5%	97%	**Governing constraint**

Table 1 — PDS verification summary. *The study's value is exactly this asymmetry: it confirms the energetics and spin design, and it flags frame stability — not the launch mechanism — as the make-or-break problem.*

THE GOVERNING CONSTRAINT

Frame stability, not the launch, decides the design

Tip-over and sliding fail in the large majority of sampled configurations, so a naive compact frame would walk or topple under launch reaction. This is the headline engineering insight: the analysis is telling me where to spend design effort. One-at-a-time tornado sensitivities identify the drivers — **exit velocity, frame weight, base width, leg travel and launch height** — all of which are base/leg geometry and mass, i.e. squarely within the designer's control.

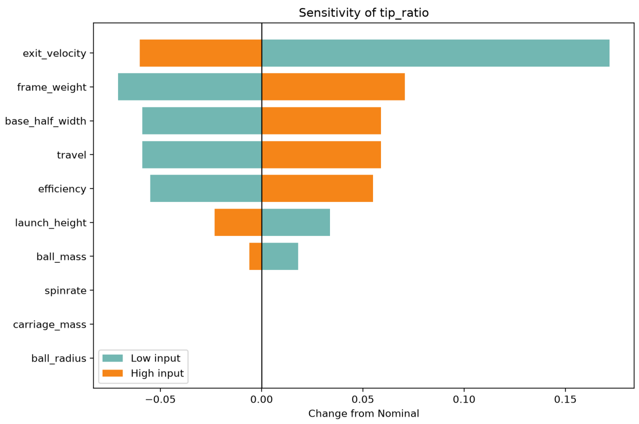

*Fig. 4 — **Tip-over sensitivity.** Exit velocity dominates; frame weight and base half-width follow.*

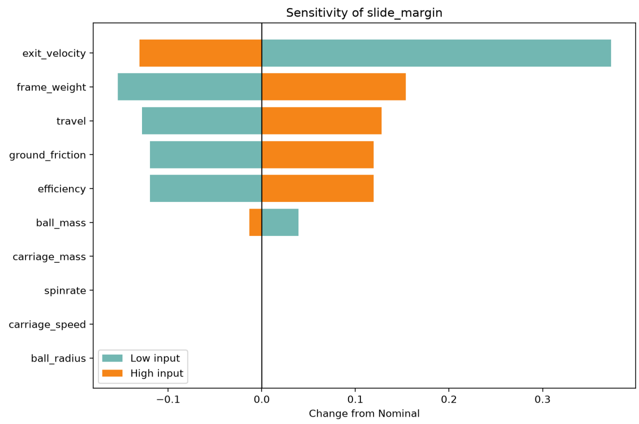

*Fig. 5 — **Sliding sensitivity.** Same levers plus ground friction; both failure modes share drivers.*

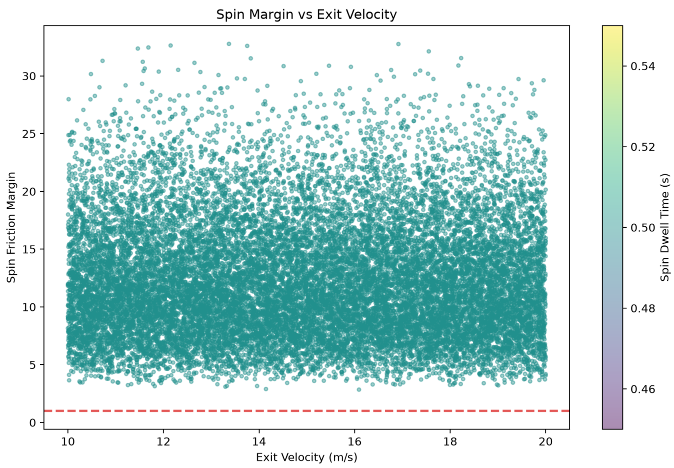

*Fig. 6 — **Spin margin vs exit velocity.** Every point sits well above the failure line (red) across the full 10–20 m/s range — the spin subsystem is robust, isolating stability as the sole governing issue.*

DESIGN RESPONSE

The result rewrote the CAD priority list. The current focus is the **base / leg stability subassembly**, and a new requirement was added — **REQ-25: foldable-but-rigid legs with a deployed stability factor** ≥ **2.0** — keeping portability while removing the tip/slide failure. Analysis is driving the design, not rubber-stamping it.

Torsional-Vibration Control of a PMSM Aircraft Drive

BEng dissertation · MATLAB / Simulink · simulation study (no hardware)

I modelled a **1.2 MW PMSM** aircraft-propulsion drive (d–q field-oriented control with inverter limits) coupled to a **two-inertia flexible drivetrain** with an ~80 Hz torsional mode, then benchmarked a velocity-feedback damper against a constrained Model Predictive Controller for vibration suppression.

The plant spans the same electrical–mechanical boundary as the launcher: an electrical drive delivering torque into a compliant structure, where control authority and actuator limits — not just the control law — decide the outcome.

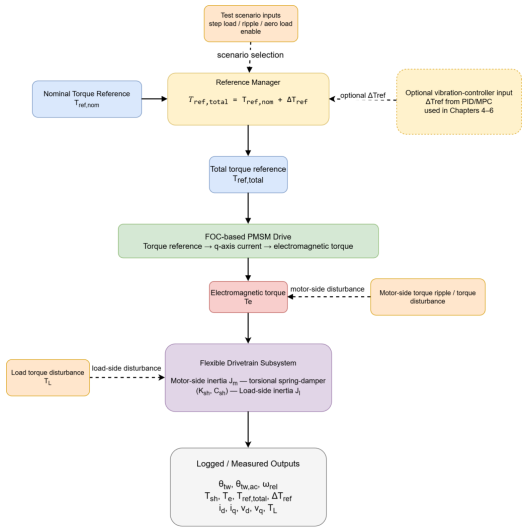

__Fig. 7 — Coupled model architecture.__ Reference manager, FOC PMSM drive, two-inertia drivetrain, logged outputs.

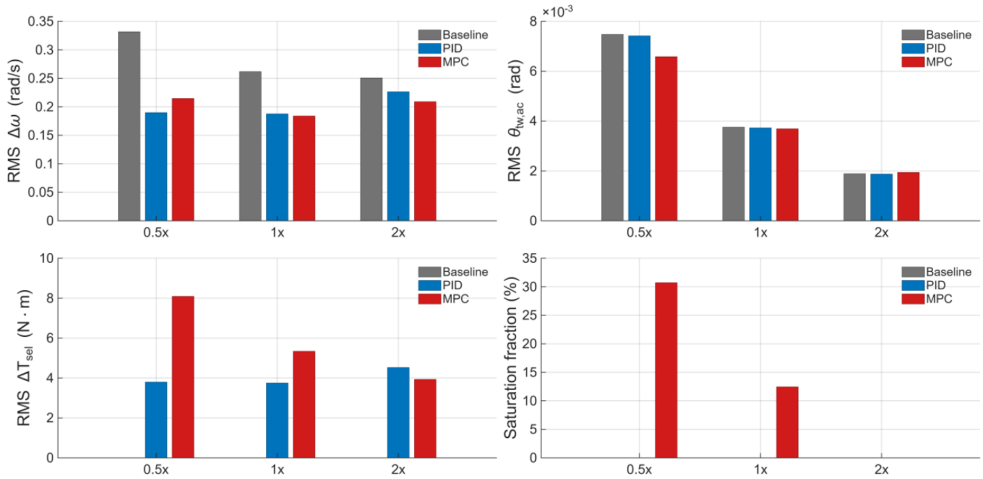

__Fig. 8 — Scenario D1 (200 Hz off-resonance) across shaft stiffness.__ RMS relative speed, twist, control effort and saturation for uncontrolled baseline vs PID damper vs MPC. Neither controller wins every metric.

KEY FINDING — actuator authority is the binding constraint

D1 RMS relative-speed reduction: 0.5× stiffness PID 42.8% / MPC 35.3% · 1× 28.4% / 29.8% · 2× 9.7% / 16.7%. MPC only pulls ahead once torque headroom exists. **R1 (80 Hz, near-resonance):** uncontrolled vibration amplified 12.8×; both controllers saturate ~85% of the window and land at near-parity (PID 56.5% / MPC 52.4%). The conclusion: under saturation, **actuator authority — not controller architecture — sets the limit.**

TRUTH-SEEKING / VERIFICATION

A deliberate verification workflow caught and corrected **two silent bugs** before final results: an inverted controller-switch polarity, and an unscaled MPC cost function. Both produced plausible-looking plots while being wrong — a reminder that a result is only as good as the checks behind it.

IN CLOSING

Two projects, one way of working

Both projects start from the same conviction: an honest, quantified model should decide a design before money is spent building it. The launcher's feasibility study earned its keep by **finding a weakness** — frame stability — rather than confirming a hope, and that result is now steering the CAD. The dissertation's numbers only became trustworthy after a verification pass **caught two silent bugs** that had produced convincing but wrong plots. In both, the useful output was not just a figure but a clear-eyed statement of what the figure does and does not prove.

I am most comfortable at the mechanical–electrical boundary, taking a system from stakeholder needs through requirements and first-order analysis to the point where the numbers — and their limits — are clear, and the next design move is obvious. That is the way of working I would bring to Amodo's hardware.

	01 · Football Launcher	02 · PMSM Vibration Control
Discipline	Mechanical design & systems engineering	Control & power electronics
Toolset	Python · Monte-Carlo · sensitivity analysis	MATLAB / Simulink · d–q FOC · PID / MPC
Scope	Concept → 25-req PDS → 20,000-run study	1.2 MW drive + two-inertia drivetrain model
Headline insight	**Frame stability** is the governing constraint — found by analysis, now driving CAD	Under saturation, **actuator authority** — not controller type — is the limit
Status	Analysis complete; CAD & prototype next	Dissertation complete (simulation only)

Demonstrated in these projects
Requirements & systems engineering (V-model, 25-requirement PDS) · first-order mechanical sizing (energy, force, stability) · Monte-Carlo feasibility & sensitivity analysis · Python (NumPy / pandas / matplotlib) · control & power-electronics modelling (d–q FOC, PID, MPC) in MATLAB/Simulink · disciplined verification.

Launcher — next phase
Parametric CAD of the stability-driven frame · fully-defined GD&T drawings · design for manufacture · physical prototype and test against the PDS pass/fail rules. Progress is pushed to the live repository as it lands.

LIVE PROJECT REPOSITORY

This PDF is a **snapshot of the launcher at application stage**. The live repository holds the latest version of the project, where I am publishing the CAD, engineering drawings, Python feasibility model, results, manufacturing notes and prototype evidence **as the work progresses**:
github.com/ChristianCI05/Football-launcher-design-portfolio

Every figure and number in this portfolio comes from my own models — I'd be happy to walk through the assumptions, calculations or code behind any of them.

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